

Homework 2

Started: Sep 24 at 11:23am

Quiz Instructions

Note that homework assignments are **not** automatically submitted on the due date/time. You must submit them yourself once you are done.



Question 1 2 pts

The best way to ask the instructor a question outside of the classroom and office hours is:

☐

by email

☐

by phone

☐

by postal service

☐

via Canvas



Question 2 2 pts

MPI stands for "Message Passing Interface".

☐

True

☐

False



Question 3 2 pts

A cyclic distribution of the elements in an array is useful for load balancing when the amount of work per element decreases with increasing array indices.

☐

True

☐

False



Question 4 2 pts

A cyclic distribution of the elements in an array is useful for load balancing when the amount of work per element decreases with increasing array indices.

☐

True

☐

False



Question 5 2 pts

When parallelized, the Collatz code from the projects is likely to suffer from load imbalance.

☐

True

☐

False



Question 6 2 pts

The serial fractal code from the projects contains a loop-carried data dependence in the innermost loop (that iterates over the columns).

☐

True

☐

False



Question 7 2 pts

MPI programs must be run with at least two processes.

☐

True

☐

False



Question 8 2 pts

MPI programs cannot suffer from indeterminacy.

☐

True

☐

False



Question 9 2 pts

In this course, we should always call MPI_Init with two NULL parameters.



True



False



Question 10 2 pts

A single call to `MPI_Reduce` by each process suffices to reduce local histograms with k buckets into a global histogram with k buckets.



True



False



Question 11 2 pts

In `MPI_Reduce`, every process must pass a parameter for the destination buffer, even processes that will not receive the result of the reduction.



True



False



Question 12 2 pts

In an MPI program with 16 processes, what is the smallest rank that any process will have?



Question 13 2 pts

In an MPI program with 16 processes, what is the largest rank that any process will have?



Question 14 2 pts

When running an MPI program with 16 processes that call `MPI_Gather` using the default communicator, how many processes will receive data?



Question 15 2 pts

When running an MPI program with 16 processes that call MPI_Scatter using the default communicator, how many processes will receive data?



Question 16 2 pts

When running an MPI program with 16 processes that call MPI_Gather using the default communicator where each source process sends an array of 3 elements, how many elements does the destination process receive?



Question 17 2 pts

When running an MPI program with 16 processes that call MPI_Reduce using the default communicator where each source process sends an array of 3 elements, how many elements does the destination process receive?



Question 18 2 pts

When running an MPI program with 16 processes that call MPI_Bcast using the default communicator where the source process sends an array of 3 elements, how many elements does each destination process receive?



Question 19 2 pts

When running an MPI program with 16 processes that call MPI_Scatter using the default communicator where the source process scatters an array of 64 elements in total, how many elements does each destination process receive?



Question 20 2 pts

The receive buffer size parameter in MPI_Recv calls specifies the exact length of the message to be received (in number of elements).

☐

True

☐

False



Question 21 2 pts

MPI_Ssend may return before the receiver has started receiving the message.

☐

True

☐

False



Question 22 2 pts

MPI_Recv may return before the message has actually been received.

☐

True

☐

False



Question 23 2 pts

MPI_Reduce implies a barrier.

☐

True

☐

False



Question 24 2 pts

MPI_Allgather implies a barrier.

☐

True

☐

False



Question 25 2 pts

Returning from an MPI_Reduce call by any process implies that the process receiving the reduced result has already reached its MPI_Reduce call.

☐

True

☐

False



Question 26 2 pts

All collective communication MPI functions we have discussed in the lectures imply some synchronization.

☐

True

☐

False



Question 27 2 pts

MPI_Allgather can be emulated with MPI_Gather followed by MPI_Bcast.

☐

True

☐

False



Question 28 2 pts

MPI_Allreduce is typically faster than calling MPI_Reduce followed by MPI_Bcast.

☐

True

☐

False



Question 29 2 pts

MPI_Reduce has a similar communication pattern (sends and receives of messages) as MPI_Gather.

☐

True

☐

False



Question 30 2 pts

MPI_Allgather has a similar communication pattern (sends and receives of messages) as MPI_Allreduce.

☐

True

☐

False



Question 31 2 pts

The communication pattern (one-to-one, one-to-many, many-to-one, or many-to-many) of MPI_Scatter and MPI_Bcast are identical.

☐

True

☐

False



Question 32 2 pts

MPI_Scatter performs one-to-many communication.

☐

True

☐

False



Question 33 2 pts

MPI_Gather performs one-to-many communication.

☐

True

☐

False



Question 34 2 pts

MPI_Send performs one-to-many communication.

☐

True

☐

False



Question 35 2 pts

MPI_Reduce performs one-to-many communication.

☐

True

☐

False



Question 36 2 pts

MPI_Bcast performs one-to-many communication.

☐

True

☐

False



Question 37 2 pts

MPI_Allgather performs one-to-many communication.

☐

True

☐

False



Question 38 2 pts

MPI_Allreduce performs one-to-many communication.

☐

True

☐

False



Question 39 2 pts

After which of the following MPI calls does the sender know that the receiver has started receiving the message? Check all that apply.

☐

MPI_Bcast

☐

MPI_Reduce

☐

MPI_Scatter

☐

MPI_Send



MPI_SSend



Question 40 2 pts

In all collective communication MPI functions discussed in class, all ranks are both senders and receivers.



True



False



Question 41 2 pts

In the call `MPI_Reduce(a, b, c, MPI_DOUBLE, MPI_MIN, 2, MPI_COMM_WORLD)`, the result is written into the "c" array.



True



False



Question 42 2 pts

In the call `MPI_Reduce(a, b, c, MPI_DOUBLE, MPI_MIN, 2, MPI_COMM_WORLD)`, process b is the destination.



True



False



Question 43 2 pts

In the call `MPI_Reduce(a, b, c, MPI_DOUBLE, MPI_MIN, 2, MPI_COMM_WORLD)`, each process contributes 2 elements to the reduction.



True



False



Question 44 2 pts

The call `MPI_Reduce(a, b, c, MPI_DOUBLE, MPI_MIN, 2, MPI_COMM_WORLD)` performs a sum reduction.



True



False



Question 45 2 pts

The MPI_Gather function concatenates the data from all involved processes by rank order.



True



False



Question 46 2 pts

MPI_Gather may be non-blocking on more than one of the involved processes.



True



False



Question 47 2 pts

In MPI_Gather, the chunk from rank 0 always ends up at the beginning of the final result.



True



False



Question 48 2 pts

If each process calls MPI_Reduce $k > 1$ times with the same destination, operator, and communicator, the reductions are matched based on the order in which they are called.



True



False



Question 49 2 pts

MPI_Aint denotes an integer type that is large enough to hold an address.



True



False



Question 50 2 pts

Sending one long message in MPI is typically more efficient than sending multiple short messages with the same total amount of data.



True



False

Not saved

Submit Quiz